•[1]:
```python
my_list = [10, 3.5, "hello", True]

print("List:", my_list)
print("Type of list:", type(my_list))

for i in my_list:
    print(i, "->", type(i))
```

```
List: [10, 3.5, 'hello', True]
Type of list: <class 'list'>
10 -> <class 'int'>
3.5 -> <class 'float'>
hello -> <class 'str'>
True -> <class 'bool'>
```

[2]:
```python
a = [1, 2, 3]
b = [4, 5, 6]
print("Concatenation:", a + b)
print("Repetition:", a * 3)
print("Is 2 in list a?", 2 in a)
print("Length of list a:", len(a))
print("Length of list b:", len(b))
```

```
Concatenation: [1, 2, 3, 4, 5, 6]
Repetition: [1, 2, 3, 1, 2, 3, 1, 2, 3]
Is 2 in list a? True
Length of list a: 3
Length of list b: 3
```

[5]:
```python
nums = [10, 5, 20, 8, 15]
print("Maximum value:", max(nums))
print("Minimum value:", min(nums))
print("Sum:", sum(nums))
nums.sort()
print("Ascending:", nums)
nums.sort(reverse=True)
print("Descending:", nums)
print("Index of 8:", nums.index(8))
```

```
Maximum value: 20
Minimum value: 5
Sum: 58
Ascending: [5, 8, 10, 15, 20]
Descending: [20, 15, 10, 8, 5]
Index of 8: 3
```

[6]:
```python
lst = [10, 20, 30]
lst.append(40)
print("After append:", lst)
lst.insert(1, 15)
print("After insert:", lst)
lst.remove(20)
print("After remove:", lst)
lst.pop()
print("After pop:", lst)
lst.reverse()
print("After reverse:", lst)
```

```
After append: [10, 20, 30, 40]
After insert: [10, 15, 20, 30, 40]
After remove: [10, 15, 30, 40]
After pop: [10, 15, 30]
After reverse: [30, 15, 10]
```

[8]:

```python
nums = list(map(int, input("Enter numbers separated by space: ").split()))
even = []
odd = []
for i in nums:
    if i % 2 == 0:
        even.append(i)
    else:
        odd.append(i)

print("Even numbers:", even)
print("Odd numbers:", odd)

print("Largest number:", max(nums))
print("Smallest number:", min(nums))
unique_nums = list(set(nums))
print("Without duplicates:", unique_nums)
```

```
Enter numbers separated by space:  57
Even numbers: []
Odd numbers: [57]
Largest number: 57
Smallest number: 57
Without duplicates: [57]
```